

**Independent Authoring****A01 / GATE 5 REFERENCE**

# Another author built a working domain without acquiring admission authority

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**HOW + WHAT**

→ External author

→ Independent exam

→ Owner admission

DESIGN TIME / Authoring and execution are separate processes.

Gate 5 asks whether Radiolaria's way of constructing governed work can be transferred to a developer who did not build its core. The accepted example is a football booking domain written by an external model author. It computes offers from a synthetic venue's inventory, exchanges the result between two independent owners, performs a separately authorized mock booking and preserves verifiable history. The authoring result connects the Football and External DRS views: the program that performs those tasks has an inspectable construction history of its own.

The engineering basis of this dossier is commit 64f2b38b0d671e4907bae8649fff8f3d2ee0751c. Functional acceptance followed an explicitly amended authoring protocol: an initial submission and three corrections remained INCOMPLETE; one separately authorized fourth correction completed the technical acceptance. The final program was then reviewed and admitted through a different owner process. A successful submission did not install itself.

Here, Root means the local authority that decides for its own system. Work is the native unit of computation; Host and Firewall bind permitted execution and effects. The author received mechanisms for connecting a domain to these interfaces, not the power to overrule them. This is a finite transfer result for a reviewed Python reference profile. It supports the architectural interpretation that construction knowledge can travel while admission authority stays with the receiving owner. It does not establish unrestricted automatic software creation or an integrated synthesis factory.

**Sources:** [A-S-d1result](#); [A-S-landing](#); [A-S-contract](#); [Full exact source map](#).

# HOW specifies construction; WHAT specifies the football task

The public kit separates architecture from the requested application. HOW\_TO\_BUILD describes native Work, local Root review, pointer-first DRS retrieval, source provenance, currentness and effect admission. Its machine interface map gives exact callable signatures, fields, file locations and hashes extracted from the supplied source. The approved snapshot is a read-only interface reference. It is not a second editable kernel or a permission to use every function found in that snapshot.

The generic numerical example computes  $3 \times x + 2$ , giving 23 for input 7 and 35 for input 11. A distinct two-record MOCK example demonstrates bounded effect and receipt behavior. Neither example supplies a pitch-selection algorithm. The separate WHAT\_TO\_BUILD brief introduces the football conditions, finite inventory, half-open intervals, integer prices, explicit consent, deterministic preference order and evidence obligations. Knowing that a natural full-size field and ninety-minute sessions are required is ordinary task disclosure; handing over a winning set of slots would be a different intervention.

The football evidence fields are explicitly labeled as a task-specific profile in CANDIDATE\_OUTPUT and FOOTBALL\_PROFILE. Their names are not universal Radiolaria primitives. The shared  $E \rightarrow B \rightarrow C$  contract explains how an original computed result becomes published evidence and then verified consumption context. A future author can reuse that construction contract with another supported WHAT, subject to independent review. The kit therefore teaches a bounded method with concrete interfaces, rather than promising any program from an unconstrained natural-language instruction.

| Input                              | What it contributes                                                |
|------------------------------------|--------------------------------------------------------------------|
| HOW + machine interfaces           | Construction rules, signatures, hashes and native seams            |
| Approved source + generic examples | Read-only mechanisms; numerical and two-record MOCK demonstrations |
| WHAT + FOOTBALL_PROFILE            | Business constraints and task-specific evidence mapping            |
| Private examiner                   | Outside the author input; independent acceptance machinery         |

Sources: [A-S-how](#); [A-S-what](#); [A-S-interface](#); [A-S-output](#); [Full exact source map](#).

A03 / GATE 5 REFERENCE

# The broker exposed seven bounded tools; execution belonged to the reviewer

The recorded author used gpt-6-astra with literal ultra effort and the default service tier. Its seven advertised broker tools could read or search approved sources, read candidate files, write or patch candidate text, inspect syntax as data and freeze a submission. A patch required the expected whole-file SHA-256 and a unique literal replacement. Submission caused the broker to calculate the package manifest. These operations provide a concrete account of what the author could see and change.

Source review and execution were separate stages. The supervisor read the initial candidate and every production change before approving a bounded runner invocation. The trusted parent created publisher and requester processes, assigned their role-specific inputs and channels, and recorded process results. The private examiner used independent source inputs and observations, rather than treating candidate-authored counters as an oracle. Later repository admission was another operation owned outside the candidate.

The white-room description is a metaphor for this declared access surface. It does not claim absence of pretraining, all platform context or all prior knowledge. The author broker is not a universal sandbox for arbitrary hostile Python. The reviewed-reference trust base includes the interpreter, approved source, runner, observer, independently frozen inputs and examiner. Same-process observation and shared worker UID are declared limitations. The earlier standalone command/exec canary discrepancy remains historical evidence; success through the narrower broker did not retroactively repair that different route.

| Broker tool                                   | Permitted operation                              |
|-----------------------------------------------|--------------------------------------------------|
| read_approved_source / search_approved_source | Pinned public text only                          |
| read_candidate_file                           | Candidate bytes and whole-file hash              |
| write_candidate_file / patch_candidate_file   | Inert candidate edits; exact patch precondition  |
| inspect_candidate_text                        | Parse Python / JSON without importing or testing |
| submit_candidate                              | Freeze package; broker generates candidate.json  |

Sources: [A-S-tools](#); [A-S-execution](#); [A-S-authorreceipt](#); [A-S-dreport](#); [Full exact source map](#).

A04 / GATE 5 REFERENCE

# The chronology preserves transport failures, incomplete trials and the accepted continuation

The history contains different kinds of events, so counting every stage as another failed author would obscure what occurred. The initial G54 Gemini transport received no response from two generation POST attempts; provider completion remained unknown and there was no submitted candidate. G54R recovered the transport and completed an initial submission plus three corrections with gemini-2.5-flash, but its final candidate failed before native Work and Root. The retained result identifies an unsupported invalidation\_contexts constructor argument.

G54B investigated an Astra access route. Synthetic command/exec controls did not establish the required boundary, so no football task was delivered and no author trial started. G54B1 then used the no-environment broker route: an initial submission and three corrections reached useful two-owner computation and a mock effect, but the independent acceptance stopped at venue\_offer\_consumption. The separately published B1 account remains an INCOMPLETE historical result.

The C-series prepared and reviewed a corrected contract and examiner; it did not consist of new blind football authors. G54D was a new cold trial on the revised kit. Its initial submission and three corrections reached 15 of 16 native cases. G54D1 continued that same development lineage under an explicit extra correction and completed the technical checks. G55L subsequently admitted and published the engineering result. G56 documents that history with its own publication status; it does not rewrite earlier outcomes.

| Stage      | Recorded outcome                                                            |
|------------|-----------------------------------------------------------------------------|
| G54 / G54R | Gemini transport failure, then recovered initial + 3; INCOMPLETE            |
| G54B       | Boundary probe; no author task delivered                                    |
| G54B1      | Astra initial + 3; useful MOCK; contract conflict; INCOMPLETE               |
| G54C-C4    | Contract / examiner preparation and finite reviewed-reference qualification |
| G54D       | New Astra cold trial, initial + 3; 15/16; INCOMPLETE                        |
| G54D1      | Restored continuation, correction 4 / submission 5; accepted                |
| G55L / G56 | Engineering published / this separate publication stage                     |

Sources: [A-S-g54initial](#); [A-S-history](#); [A-S-dreport](#); [A-S-amendment](#); [Full exact source map](#).

## A05 / GATE 5 REFERENCE

# The earlier contract failure tested the instructions as well as the candidate

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Producer result E      → Later publication B      → Checked context C      → Independent joins

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CONTRACT MECHANISM / The old B1 examiner confused distinct representations.

G54B1 revealed an important distinction between a computation and the evidence later built around it. E is the producer Work's actual result. B publishes that result together with the identity of the completed Work and its later review. C adds the receiving side's validated status and request context. They belong to one causal history, but they cannot be identical files: the original computation cannot contain identifiers of consequences that do not yet exist.

The old examiner required the whole published body at both Work boundaries while the public instruction required that body to be assembled after producer Work and review. The revised general contract checks the causal joins instead: `payload_projection(B)` equals E; source and review references resolve to actual evidence; expected C is independently derived from the signed body, current authenticated status, original request and local policy. The consumer's own input cannot be recycled as the expected input, since that would merely compare a claim with itself.

The correction was expressed generically and exercised with a numerical profile before the next cold author. The final football profile makes its mapping explicit: `B.offer` equals E, `C.body` equals B, and the consumer receives canonical C. This is an informative limit of instruction transfer. An independent author can expose a defect in the demanded proof as well as in its own implementation. The historical B1 failure remains recorded; a versioned repair to the shared contract supports the later success. Frozen examination is valuable because changes are visible, not because an examiner is assumed infallible.

**Sources:** [A-S-history](#); [A-S-b1review](#); [A-S-shared](#); [A-S-output](#); [Full exact source map](#).

A06 / GATE 5 REFERENCE

# G54D stopped at a genuine missing decision after the correction budget ended

The revised-kit cold trial was a real development sequence. Its initial submission failed at the Root binding used for native DRS descent. The first correction reached the offer and consumption path, then failed before effect dispatch on a capability operation/namespace mismatch. The second correction performed the main booking and history story but left a revoked publisher report and a generated source reference incompatible with the existing DRS filter. Those were concrete candidate defects, not evidence that every refusal came from a bad examiner.

The third correction fixed the issues actually reported to the author. The final matrix then exposed one remaining failure: a verified REVOKED status correctly stopped use and produced REFUSED without a booking, but the known exception branch had never invoked a local Root refusal decision. Earlier metadata and retrieval ACCEPT decisions did not provide that missing decision. The program's stop was safe in this observed case; it still failed the required evidence obligation.

The absent Root call existed before the last correction. The earlier reviewer stopped at other failures, so the specific omission was not delivered before that final permitted attempt. The process therefore could not honestly be summarized as three unsuccessful attempts to fix the same diagnosed error. The original limit was exhausted and G54D stayed INCOMPLETE. The subsequent decision was to clarify the profile's refusal obligation and explicitly allow one additional authored correction, preserving both the useful result and the failed protocol outcome.

| Submission   | Feedback / final finding                                     |
|--------------|--------------------------------------------------------------|
| Initial      | Invalid Root binding in native DRS descent                   |
| Correction 1 | Native capability operation / namespace mismatch             |
| Correction 2 | Revoked publisher report and source-reference representation |
| Correction 3 | Earlier items repaired; missing refusal Root first exposed   |

Sources: [A-S-dreport](#); [A-S-restoration](#); [A-S-clarification](#); [A-S-amendment](#); [Full exact source map](#).

A07 / GATE 5 REFERENCE

# The fourth correction was a restored continuation, not a new blind trial

The amendment allowed exactly one additional correction by the author, index 4 and submission 5. The supported attempt to resume the old ephemeral thread returned no rollout found. Rather than invent continuity, the record declared RESTORED\_CONTINUATION and preserved the failed resume response. The new model turn retained gpt-6-astra, ultra and default settings, but identical hidden model state was neither recovered nor claimed.

The restoration inventory names precisely what was supplied: the same public architecture/task input, the author's own final source, the feedback already delivered during G54D, and a new narrowly authorized explanation of the missing refusal. It omitted private examiner code, private case inputs, expected answers, old B1 solution code and any ready-made repair patch. The full earlier tool transcript and opaque model state were not reconstructed. The twenty-two seeded author files and their hashes identify the program being continued.

One new transport turn performed seventy-nine broker operations and submitted one corrected pack. Tool operations are reads, searches, edits, inspection and submission; they are not seventy-nine model requests. The distinction matters because the accepted claim is continuity of the recorded development lineage and exact code provenance, not a fresh test of total ignorance. The historical final reserved variation also remained the preserved G54D challenge. Its reuse after continuation supports regression and bounded variation evidence; it is not presented as a newly unseen independent benchmark.

| Restored                                       | Explicitly not restored / supplied                |
|------------------------------------------------|---------------------------------------------------|
| Exact public kit and original public task      | Opaque original model state or hidden reasoning   |
| Author's own final G54D source                 | Old B1 target implementation or ready-made repair |
| Previously delivered feedback + new diagnostic | Private examiner, fixtures or expected selections |

Sources: [A-S-amendment](#); [A-S-restoration](#); [A-S-resume](#); [A-S-authorreceipt](#); [Full exact source map](#).

## A08 / GATE 5 REFERENCE

# The seed and recorded edits reconstruct the accepted source bytes

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22-file seed → 79 broker operations → 23 authored files → 24-file package

RECORDED CONSTRUCTION / The additional file is the broker manifest.

Authorship is supported by a reproducible relationship between inputs and outputs. The reconstruction began with twenty-two exact author files from the final G54D submission. It applied the recorded broker edits in order and independently compared the resulting files with the submitted pack. This produced twenty-three author files. The broker separately generated candidate.json from the file inventory, bringing the complete package to twenty-four files. The manifest is coordinator output and is not falsely credited as model-written application logic.

The recorded changes were narrow. The production delta was in candidate\_domain/run.py. The author added tests/test\_status\_refusal.py and updated the passport, README and source-impact explanation. Existing domain selection, policy, effect, transport and other application files retained their hashes. The source review read the complete production delta and surrounding requester flow. It found no new production imports, provider or process calls, observer access or cross-role reads; the added evidence writes used existing local helpers.

This is stronger attribution than an assistant saying that it wrote a program: exact final bytes can be derived from the known seed and observed author operations. It also keeps the support work visible. The supervisor provided task material, feedback and review machinery, but did not implement the candidate repair. The authored component tests were inspected as source and syntax; their presence does not establish a separate executed test result. Native acceptance comes from the independent runner evidence, not from file names that begin with test.

**Sources:** [A-S-reconstruction](#); [A-S-sourcereview](#); [A-S-patch](#); [A-S-bindings](#); [Full exact source map](#).

A09 / GATE 5 REFERENCE

# The repair obtained a real refusal and bound it to the rejected source

The repaired branch preserves authenticated terminal status before refusing further use. On the known status\_not\_active path it revalidates retained signed evidence at the recorded observation times, then invokes a fresh local Root review whose real decision is BLOCKED\_FAIL\_CLOSED. The completed refusal binds the original request, actual reason, source revision and the relevant pointer/status evidence. It does not rename an earlier ACCEPT record or write a decorative rejection label.

The supplemental independent check joins those records to observed native calls in both revoked-source events. The exact event has refusal call sequence 29; the shifted event has sequence 66. These are observer call positions, not totals of Root decisions. Both native results are BLOCKED\_FAIL\_CLOSED, both use reason status\_not\_active, and their distinct request and status hashes remain linked to the actual source records. The capture records zero booking executor starts and no registry mutation for this case.

The source change was deliberately specific. It did not replace the selection algorithm, loosen the budget, manufacture consent, change the policy recipe or alter technical examiner predicates between D and D1. Unknown exceptions remain visible. A possible failure after effect start remains an uncertain outcome requiring evidence and readback; it is not rewritten as a clean no-effect refusal. That separation explains why the final correction completes an accountability path as well as a test: the local authority actually issued the decision now cited by the result.

| Recorded event       | Observed refusal call | Decision / reason                       |
|----------------------|-----------------------|-----------------------------------------|
| EXACT revoked source | 29                    | BLOCKED_FAIL_CLOSED / status_not_active |
| SHIFT revoked source | 66                    | BLOCKED_FAIL_CLOSED / status_not_active |

Sources: [A-S-clarification](#); [A-S-sourcereview](#); [A-S-refusal](#); [A-S-native](#); [Full exact source map](#).

A10 / GATE 5 REFERENCE

# Independent acceptance checked performance and kept evidence classes separate

After the correction, the same final candidate passed all sixteen native scenarios. The recorded matrix contains fifty-six events and seven isolated mock executor starts; one start belongs to the main three-slot booking story. Changed inputs, a second team, permutation, refusal paths and continuation exercise the implementation beyond one selected output. The examiner derives expected domain values from original runner inputs and frozen rules. A producer/consumer pair agreeing on an invented value would not satisfy that independent comparison.

The later live series is a separate experiment: three actual gemini-2.5-flash requests supplied semantic proposals for EXACT, SHIFT and VERIFY, followed by five native events and one additional mock execution. The proposals were precollected before booking. The VERIFY model did not see the subsequent receipt; the program's later verification consumed the saved history and readback. Controlled owner inputs supplied confirmation and repetition. Authoring and runtime semantic participation are therefore separate model roles.

Two supplied processes then checked the frozen D1 evidence without a new candidate run, provider call, Root decision or effect. Their outputs are byte-identical at 32,702 bytes. Altered-evidence controls are pure checking of data, not additional native attacks. The scope is bounded: reviewed-reference trust, synthetic inventory and MOCK action, no arbitrary hostile Python guarantee or physical venue integration. Acceptance is substantial because each class of evidence answers its own question; adding unlike counters together would make the result less informative.

| Evidence class | Recorded scope                                                 |
|----------------|----------------------------------------------------------------|
| D1 native      | 16 cases / 56 events / 7 isolated mock executor starts         |
| D1 live        | 3 provider requests / 5 events / 1 further mock executor start |
| D1 supplied    | 2 independent processes / 32,702-byte identical output         |

Sources: [A-S-native](#); [A-S-d1result](#); [A-S-live](#); [A-S-supplied](#); [Full exact source map](#).

A11 / GATE 5 REFERENCE

# Engineering admission published the reviewed bytes through a separate owner process

A technically accepted candidate remained an external result until its source was admitted. G55 assembled an exact proposal, portable evidence and the finite acceptance matrix. External Work review checked those bytes before G55L used the unchanged repository R1 process. The actual engineering commit is 64f2b38b0d671e4907bae8649fff8f3d2ee0751c, with parent 5cef2c52fd805daedec958109caf9d1806899eb4 and tree cacb14a9574c350fb33239680abe0c3a5ede7b5e. The ordinary push and independent remote readback are recorded as completed.

The admitted change adds 212 paths while preserving all 4,367 earlier tracked files and their modes. Within those additions are thirteen shared source/test/checkpoint files and the exact twenty-four-file authored pack, plus the kit and evidence. Core, Root, Host and Firewall source were not rewritten to make the target story pass. Placing the pack in the repository does not make its manifest a runtime permission or automatically execute it.

The installed offline proof returned the expected 37,389-byte canonical result. This is a repository-level verification report, distinct from the earlier 32,702-byte D1 supplied output; their different sizes describe different report generations. Historical kit and preparation documents still contain their original PENDING labels. The current status bridge uses later landing receipts instead of editing those frozen files. Engineering acceptance and publication are complete; the separate G56 documentation publication and final closure rows must receive their own actual completion evidence.

| Admission fact                   | Value                                           |
|----------------------------------|-------------------------------------------------|
| New paths                        | 212                                             |
| Earlier paths / modes preserved  | 4,367                                           |
| Installed proof                  | 37,389 bytes; expected canonical SHA-256        |
| Engineering                      | Accepted, committed and independently read back |
| G56 documentation / full closure | Separate publication acceptance still required  |

Sources: [A-S-landing](#); [A-S-preserved](#); [A-S-contract](#); [A-S-bindings](#); [Full exact source map](#).

A12 / GATE 5 REFERENCE

# The published kit is a starting contract for another reviewed extension

A developer should begin with the current kit entry, then read HOW\_TO\_BUILD, SHARED\_EVIDENCE\_CONTRACT and the machine interface map. SOURCE\_INVENTORY identifies the supplied code and schema resource; CANDIDATE\_OUTPUT describes the bounded return package. The separate WHAT\_TO\_BUILD and FOOTBALL\_PROFILE are the demonstrated application. The accepted football pack is useful as a disclosed example, but it must not be silently included in a future blind-author input. The auditor's historical corpus and private examiner are different deliverables.

A concrete extension point is the exact run\_peer\_v01 interface: role, input\_data, state\_root, output\_root, channel and bootstrap are keyword arguments supplied by the approved runner. A reviewed local body profile supplies closed body/scope fields and local validate\_body, validate\_pointer and validate\_binding functions. Neither a remote payload nor a signature selects executable code. A new implementation needs its own source review, supported runner and separate admission; this publication conveys construction knowledge, not the original owner's credentials or installation authority.

The observed transfer result is one new working domain created through this bounded contract. The architectural interpretation is that an extension's meaning and implementation can change while the receiving system retains the decision boundaries that govern use. Reflective Improvement Radiolaria, Cross-Domain Transfer Radiolaria and an integrated Capability Synthesis Factory are future directions, explicitly deferred beyond this reference. They may build on the same separation of proposals and decisions, but they are not implemented merely because a model successfully authored this package.

| Machine anchor        | Exact supported surface                                                        |
|-----------------------|--------------------------------------------------------------------------------|
| Peer endpoint         | run_peer_v01(*, role, input_data, state_root, output_root, channel, bootstrap) |
| Reviewed body profile | Closed fields; validate_body / validate_pointer / validate_binding             |
| Package manifest      | g53.candidate.v01; installable_now=false; exact byte hashes                    |
| Future extensions     | Reflective Improvement / Cross-Domain Transfer / Factory: deferred             |

Sources: [A-S-kitstart](#); [A-S-how](#); [A-S-interface](#); [A-S-inventory](#); [Full exact source map](#).